Rural Roads and Local Economic Development

By Sam Asher and Paul Novosad (2019)

Presented by Zach Winship

Roadmap

01 Motivation

02 Background/Data

03 Methodology

04 Empirical Results

05 Robustness

06 Conclusions

01

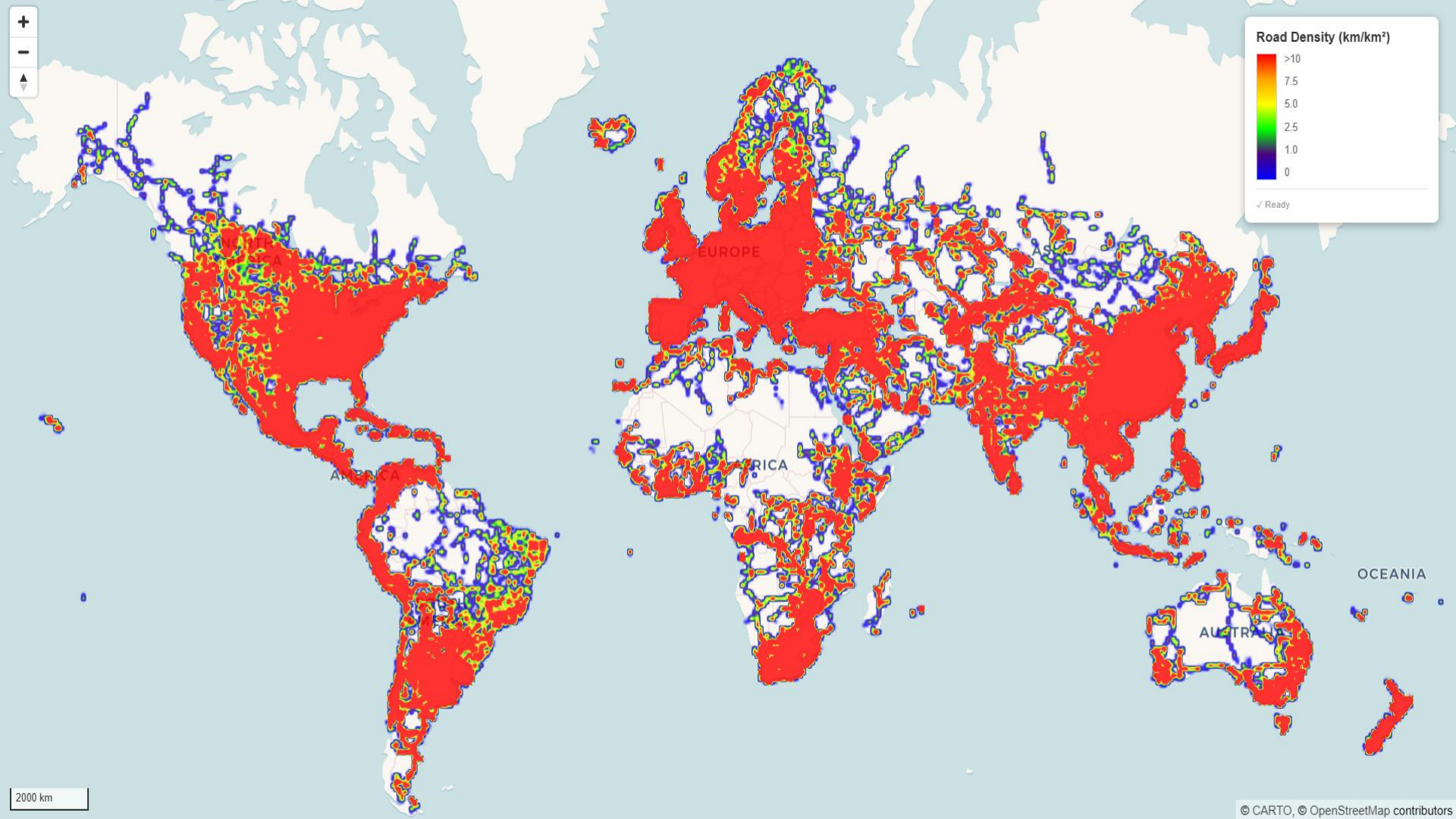
Motivation

The Global Challenge

Lack of Rural Roads

Over 1 billion people worldwide live more than 2 km from paved roads

(The World Bank Group, 2016)


Road Density (km/km²)
>10
7.5
5.0
2.5
1.0
0
✓ Ready
EUROPE
OCEANIA
2000 km
© CARTO, © OpenStreetMap contributors

The Global Challenge

Lack of Rural Roads

Over 1 billion people worldwide live more than 2 km from paved roads

Isolation

Without roads, communities are isolated from the global economy

Loss of Network Effects

- Labor specialization
- Economies of scale
- Information asymmetry

(The World Bank Group, 2016)

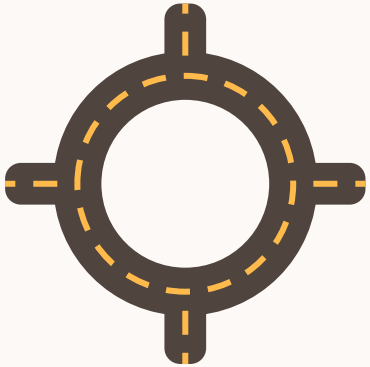
The Measurable Economics of Isolation

Transportation

Transportation costs are not linear

Time Tax

Large portions of the day are spent on “unproductive” transportation



Agriculture

Development toward modern technologies is difficult

Market Prices

Disconnect from a larger market leaves prices volatile

Berg, C., Blankespoor, B., & Spearot, A. (2025)

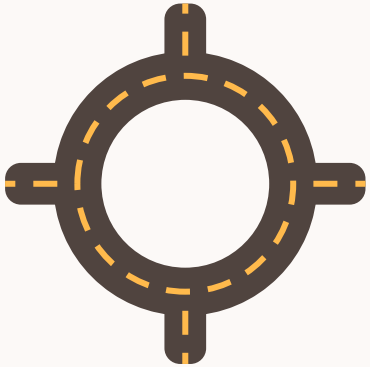
The Unrealized Economics of Isolation

Entrepreneurship

Innovation is stifled in small markets

Incentives

When markets remain constrained, incentives never emerge



Information

Isolation isn't just about moving goods, infrastructure moves information as well

Local Shocks

Local shocks lack a backbone for consumption smoothing

Berg, C., Blankespoor, B., & Spearot, A. (2025)

The “Ambiguous” Theory

Local Growth?

Road infrastructure creates lower transportation costs, which act as a subsidy for **local** production

Structural Shift?

Road infrastructure acts as an “exit ramp” for goods and labor to shift towards **external** regions

02

Background/ Data

Pradhan Mantri Gram Sadak Yojana

Policy Context: The PMGSY Program

Policy Specifics:

- $40 billion investment
- Over 200,000 villages
- Implementation from 2000-2015

Premise: “Poor road connectivity is the biggest hurdle in faster rural development”

Research Question

Do rural roads transform village economies?

Data Sources

SECC 2012

Every household/individual
Occupation, assets, income, land

Figure A2: Sample page from SECC

SECC ड्राफ्ट सूची - ग्रामीण

राज्य	जिला	तहसील	शहर/ग्राम	वार्ड कोड नंबर (केवल शहर के लिए)	गणन ब्लॉक -उप खंड
:RAJASTHAN	:Ajmer	:Ajmer	:Ajaysar	:0000	:0158_0

घरेलू संख्या	घर के प्रकार	ग्राम पंचायत	आदिम जनजाति वर्ग से है	वैधानिक रुप से छुड़ाया गया बंधुवा मजदूर	हाथ से मैला साफ़ करने वाले
: 0003	: साधारण	:-AJAYSAR	: नहीं	: नहीं	: नहीं

संख्या	नाम	मुखिया से संबंध	लिंग जन्मतिथि	पिता का नाम माता का नाम	वैवाहिक स्थिति#	व्यवसाय/ गतिविधि	अनु. जाति / जनजाति / अन्य	विकलांगता	शिक्षा
001	[redacted]	मुखिया	पुरुष 1953	[redacted]	2	मजदूर	अन्य	कोई नि:शक्तता नहीं	निरक्षर
002	[redacted]	पत्नी	स्त्री 1955	[redacted]	2	मजदूर	अन्य	कोई नि:शक्तता नहीं	निरक्षर
003	[redacted]	पुत्र	पुरुष 1989	[redacted]	1	मजदूर	अन्य	कोई नि:शक्तता नहीं	पूर्व माध्यमिक

भाग 1 विवरण : आवासीय/निवासीय				भाग 3 रोजगार और आय विशेषताओं					भाग 4 : विवरण सम्पत्तियां			भाग5 अ: भूमि स्वामित्व (एकड में)				भाग 5 ब: अन्य भूमि स्वामित्व		
मकान के दीवार की प्रमुख सामग्री #	मकान की छत की प्रमुख सामग्री #	मकान का मालिकाना हक की स्थिति	निवास के कमरों की संख्या	नियमित वेतन पाने वाला कोई परिवार का सदस्य	आयकर या वृत्ति कर दाता है	स्वयं की /संचालित ऐसी संस्था जो शासन द्वारा पंजीकृत है	परिवार के सबसे अधिक कमाने वाले सदस्य का मासिक आय	परिवार की आय का मुख्य स्रोत	रेफ्रिजरेटर	टेलिफोन / मोबाईल फोन	दो/तीन/चार पहिया या मछली पकड़ने की नाव पंजीकृत	स्वामित्व की भूमि (चास भूमि को छोड़कर)	कुल असिंचित भूमि	2 फसलों वाली सिचाई भूमि	अन्य सिंचित भूमि	यंत्रीकृत तीन/चार पहिया कृषि उपकरण	सिचाई उपकरण(ट्रलक्स , बोर , डीजल/मिट्टी के तेल/विद्युत पंप सेट. फव्वारा/ड्रिप सिचाई आदि समेत)	किसान क्रेडिट कार्ड की सीमा 50000 रुपए या अधिक है।
6	6	स्वय	4	नहीं	नहीं	नहीं	10,000 या अधिक	1	हां	केवल मोबाइल	दो पहिया	हां	1.0	3.0	1.0	नहीं	हां	नहीं

Data Sources

SECC 2012

Every household/individual
Occupation, assets, income, land

Economic Census 2013

Every non-farm firm
Employment by sector

Population Census

Demographics, amenities, services

Remote Sensing

Night lights
Agricultural vegetation

IHDS Survey

Predictions of consumption

PMGSY Admin Data

Road completion dates

Elbers et al. (2003)

Sample Overview

Sample Size

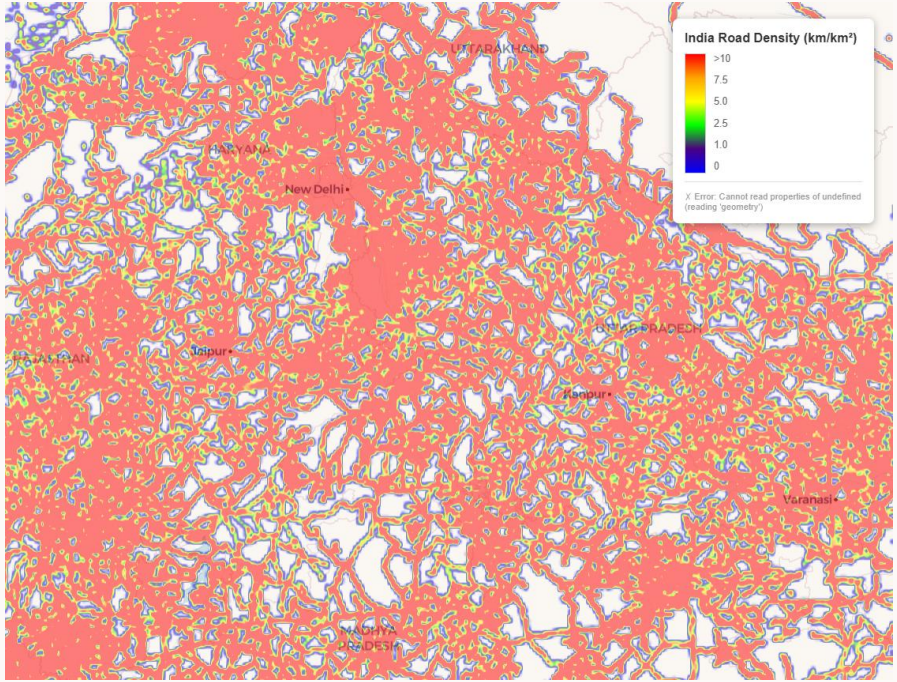
11,432 Villages

States

6 States in Sample:
Chhattisgarh, Gujarat,
Madhya Pradesh,
Maharashtra, Orissa,
Rajasthan

Timeline

Baseline: 2001
Endline: 2011-2013
Average 4 years post road



Outcomes of Interest

- Labor Allocation
  - Non-Farm Employment
- Agriculture Production & Investment
- Household Consumption & Assets

03

Methodology

Exploiting Population Thresholds

01

First Priority

Villages >1,000 in Population

02

Second Priority

Villages >500 in Population

03

Third Priority

Villages >250 in Population

Regression Discontinuity Definitions

Running Variable

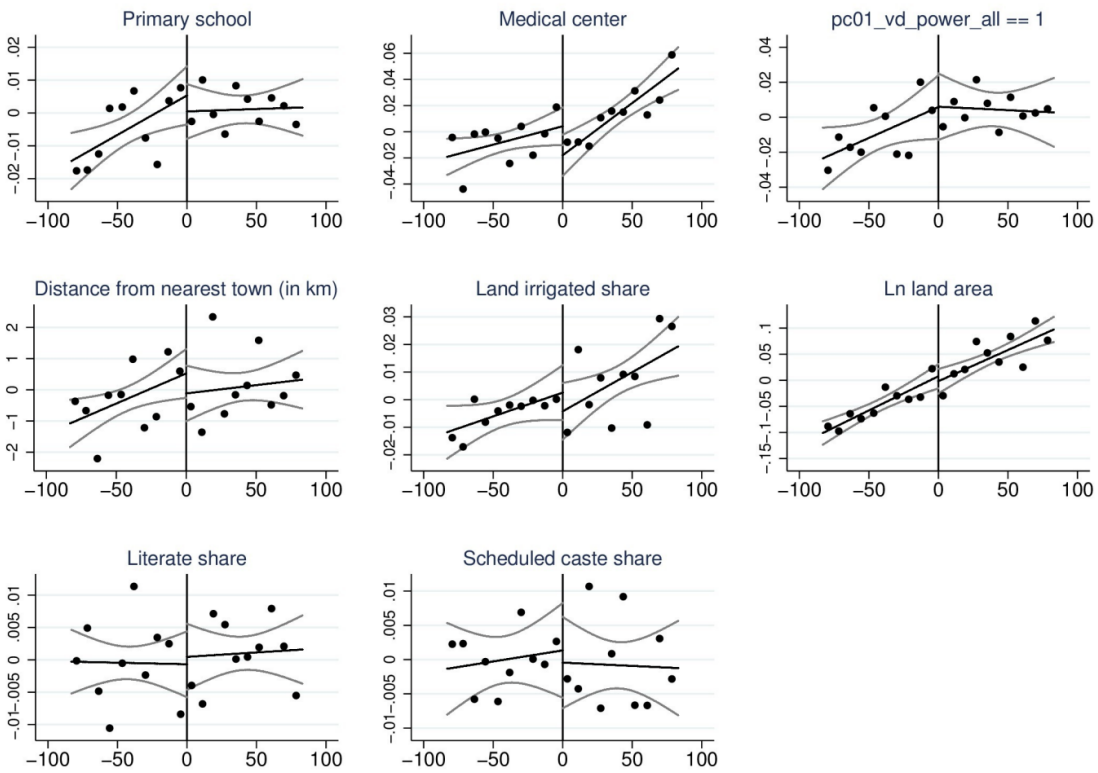
Treatment

Regression Discontinuity Assumptions

Threshold Manipulation

Balance Testing

Figure 2: Balance of baseline village characteristics



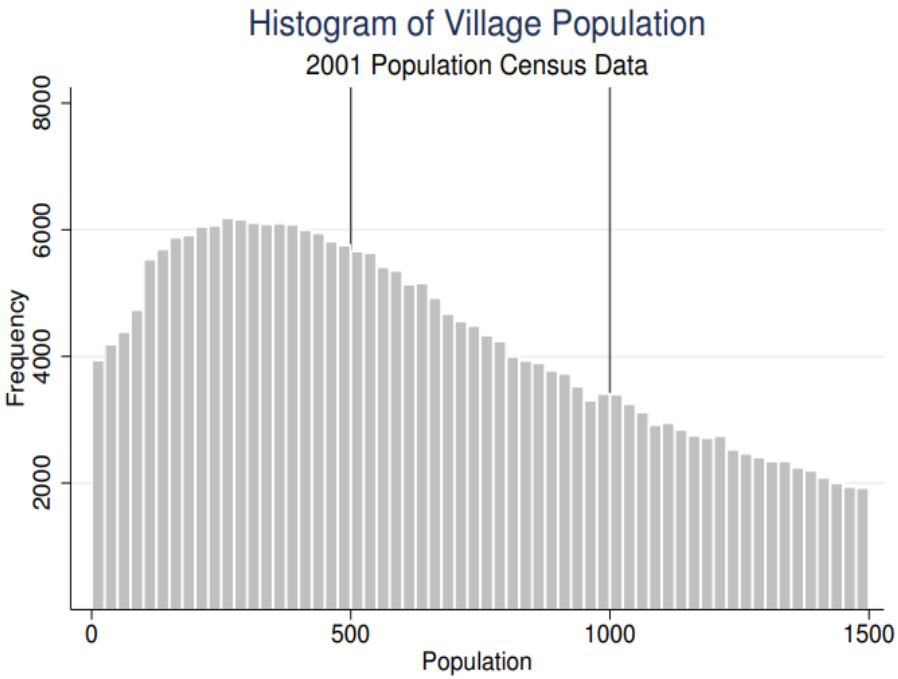
Regression Discontinuity Assumptions

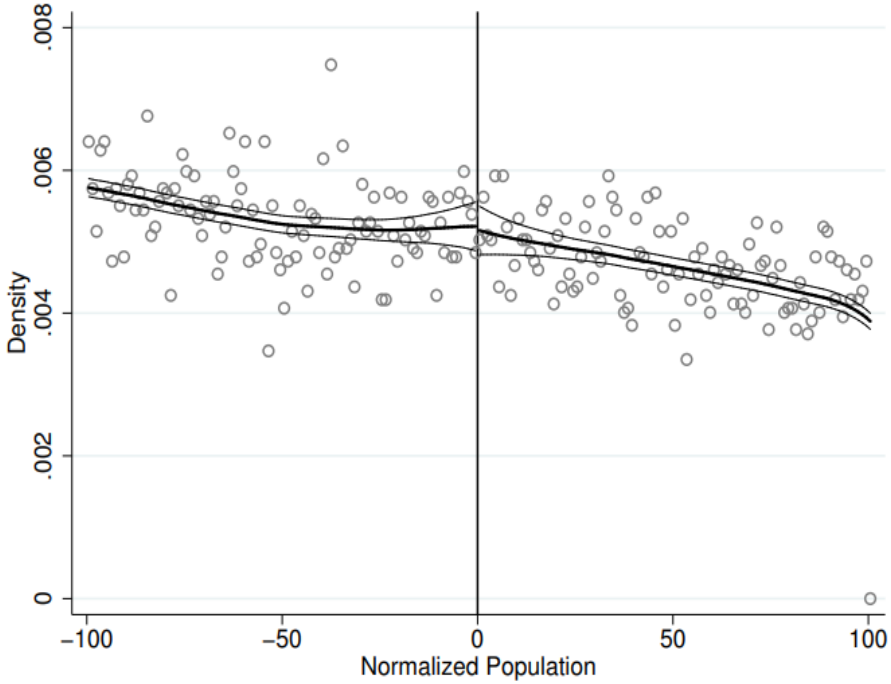
Threshold Manipulation

Balance Testing

Running Variable Distribution

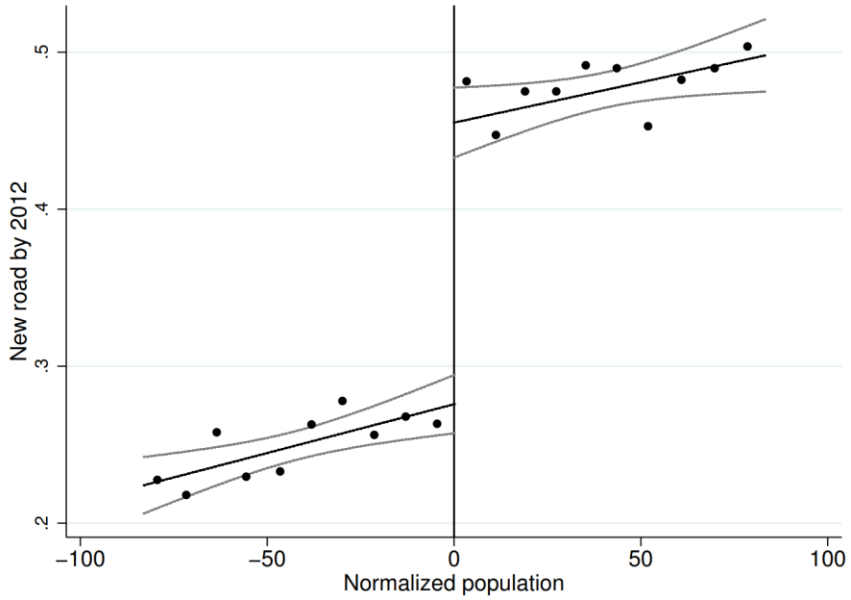
Figure 3: Distribution of running variable





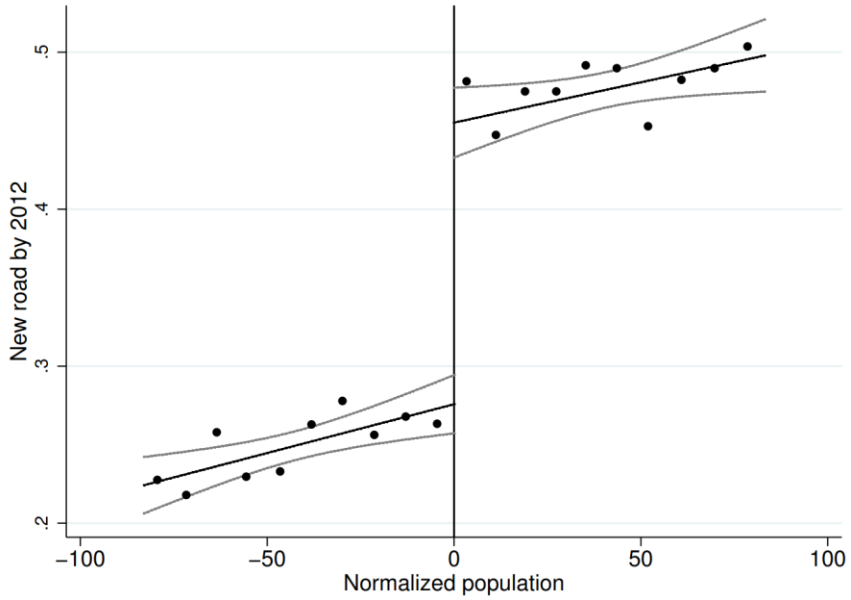
Fuzzy Regression Discontinuity Design

Figure 4: First stage: effect of road prioritization on probability of new road by 2012



Instrumenting for Road Treatment

Figure 4: First stage: effect of road prioritization on probability of new road by 2012



Regression Discontinuity Definitions

$$Running\ Variable_{v,j} = (pop_{v,j} - T)$$

$$Treatment\ Variable_{v,j} = \{pop_{v,j} \geq T\} = \begin{cases} 1 & \text{if } pop_{v,j} \geq T \\ 0 & \text{if } pop_{v,j} < T \end{cases}$$

$$Running\ Variable_{v,j} = (pop_{v,j} - T)$$

$$Treatment\ Variable_{v,j} = \{pop_{v,j} \geq T\} = \begin{cases} 1 & \text{if } pop_{v,j} \geq T \\ 0 & \text{if } pop_{v,j} < T \end{cases}$$

RD Functional Form

First Stage

$$\begin{aligned} Road_{v,j} = \gamma_0 &+ \gamma_1\{pop_{v,j} \geq T\} + \gamma_2(pop_{v,j} - T) \\ &+ \gamma_3[\{pop_{v,j} \geq T\} \times (pop_{v,j} - T)] \\ &+ \nu X_{v,j} + \mu_j + \upsilon_{v,j} \end{aligned}$$

Second Stage

$$\begin{aligned} Y_{v,j} = \beta_0 &+ \beta_1 Road_{v,j} + \beta_2(pop_{v,j} - T) \\ &+ \beta_3[\{pop_{v,j} \geq T\} \times (pop_{v,j} - T)] \\ &+ \zeta X_{v,j} + \eta_j + \epsilon_{v,j} \end{aligned}$$

04 Empirical Results

Overview of Results

Major Outcome Indices:

- Transportation Availability
- Agriculture Employment
- Number of Firms
- Agriculture Production
- Consumption

Table 3—Impact of New Road on Indices of Major Outcomes

	Transportation	Ag occupation	Firms	Ag production	Consumption
New road	0.410	−0.341	0.269	0.082	0.033
	(0.187)	(0.160)	(0.157)	(0.124)	(0.137)
p-value	0.03	0.03	0.09	0.51	0.81
Observations	11,432	11,432	10,678	11,432	11,432
R^2	0.18	0.28	0.30	0.53	0.50

Notes: This table presents regression discontinuity estimates from the main estimating equation of the effect of a new road on indices of the major outcomes in each of the five families of outcomes: transportation, occupation, firms, agriculture, and consumption. See the online Appendix for details of index construction. The specification includes baseline village-level controls for amenities and economic indicators, as well as district-cutoff fixed effects (see Section IV for details). Heteroskedasticity robust standard errors are reported below point estimates.

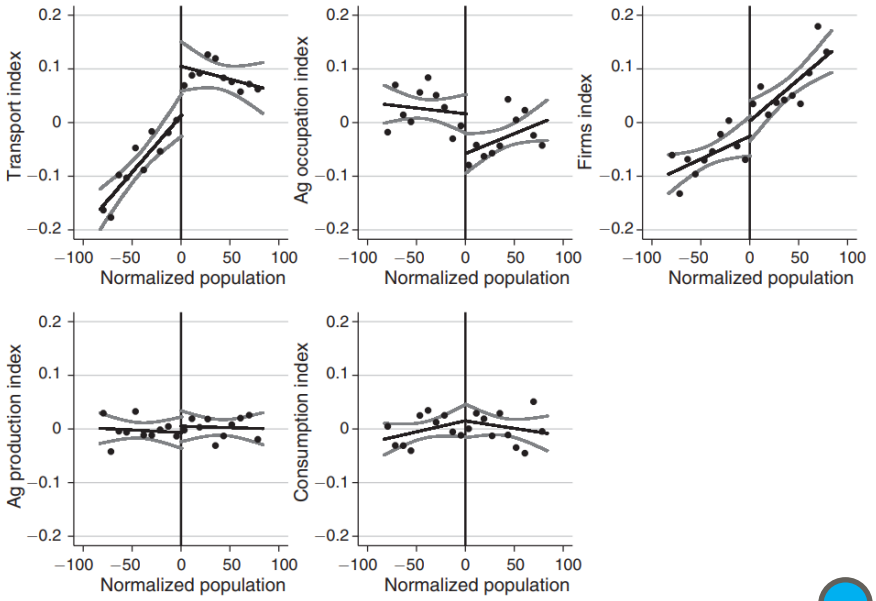


Figure 5. Reduced Form: Effect of Road Prioritization on Indices of Major Outcomes

Effects on Transportation

Table 4: Impact of new road on transportation

	Gov Bus	Private Bus	Taxi	Van	Autorickshaw
New road	0.129	0.114	0.007	-0.021	0.073
	(0.055)	(0.074)	(0.048)	(0.055)	(0.043)
Control group mean	0.118	0.205	0.069	0.156	0.055
N	11432	11432	11432	11432	11432
R2	0.30	0.10	0.09	0.44	0.26

Effects on Transportation

Transportation Type	Control Mean	Treated Mean	Treatment Effect	Standard Error
Government Bus	0.118	0.247	**0.129**	(0.055)
Private Bus	0.205	0.319	**0.114**	(0.074)
Taxi	0.069	0.076	0.007	(0.048)
Van	0.156	0.135	-0.021	(0.055)
Auto-rickshaw	0.055	0.128	**0.073**	(0.043)

Effects on Transportation

Table 4: Impact of new road on transportation

	Gov Bus	Private Bus	Taxi	Van	Autorickshaw
New road	0.129	0.114	0.007	-0.021	0.073
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N	11432	11432	11432	11432	11432
R2	0.30	0.10	0.09	0.44	0.26

Effects on Occupation and Income

Table 5: Impact of new road on occupation and income source

	Occupation		Household Income Source	
	Agriculture	Manual Labor	Agriculture	Manual Labor
New road	-0.092	0.072	-0.030	-0.011
	(0.043)	(0.043)	(0.044)	(0.044)
Control group mean	0.476	0.448	0.418	0.507
N	11432	11432	11432	11432
R2	0.28	0.26	0.31	0.28

Heterogeneity – Who's Leaving Agriculture?

Table A4: Impact of new road on distribution of landholdings

	Landless	0-2 Acres	2-4 Acres	4+ Acres
New road	-0.009	-0.012	-0.007	0.028
	(0.036)	(0.033)	(0.016)	(0.024)
Control group mean	0.434	0.287	0.120	0.159
N	11394	11394	11394	11394
R2	0.39	0.41	0.23	0.47

Effects on Village Non-Farm Employment

Table 6: Impact of new road on firms

Panel A. Log employment growth (by sector)

	Total	Livestock	Manufacturing	Education	Retail	Forestry
New road	0.273	0.252	0.260	0.198	0.333	-0.107
	(0.159)	(0.188)	(0.193)	(0.143)	(0.154)	(0.107)
N	10678	10678	10678	10678	10678	10678
R2	0.30	0.42	0.23	0.18	0.23	0.35

Panel B. Level employment growth (by sector)

	Total	Livestock	Manufacturing	Education	Retail	Forestry
New road	4.219	-1.962	2.802	0.686	1.831	2.381
	(7.596)	(3.364)	(3.794)	(0.973)	(1.534)	(4.002)
Mean employment (level)	32.1	6.9	5.8	5.1	4.5	2.8
N	10678	10678	10678	10678	10678	10678
R2	0.30	0.46	0.18	0.13	0.17	0.36

Where Did The Workers Go?

Lose Ag Workers

On average, villages lost 18.5 agriculture workers

Gain Non-Ag Workers

On average, villages gained 4.2 non-agriculture workers

Where'd They Go?

On average, 14.3 workers are unaccounted for (per village)

Where Did The Workers Go?

A

Permanent Migration

B

Commute/Temporary Migration

Where Did The Workers Go?

B

Commute/Temporary Migration

Effects on Agriculture Production

Table 7: Impact of new road on agricultural outcomes

Panel A. Agricultural yields (log)

	NDVI			EVI		
	Max - June	Cumulative	Max	Max - June	Cumulative	Max
New road	0.017	0.000	0.011	0.035	-0.001	0.022
	(0.026)	(0.013)	(0.014)	(0.033)	(0.015)	(0.019)
Control group mean	8.236	10.507	8.801	7.957	10.159	8.470
Control group SD	0.273	0.218	0.181	0.336	0.222	0.195
N	11333	11332	11333	11333	11332	11333
R2	0.71	0.89	0.82	0.72	0.86	0.72

Panel B. Agricultural inputs

	Mechanized Farm Equipment	Irrigation Equipment	Land Ownership	Non-cereal/pulse crop	Cultivated land (log)
New road	-0.004	0.002	0.006	0.030	0.040
	(0.012)	(0.028)	(0.036)	(0.073)	(0.081)
Control group mean	0.040	0.141	0.570	0.393	5.046
N	11431	11432	11432	8272	11165
R2	0.26	0.43	0.39	0.45	0.73

Effects on Consumption/Assets

Table 8: Impact of new road on predicted consumption, earnings and assets

Panel A. Consumption and earnings

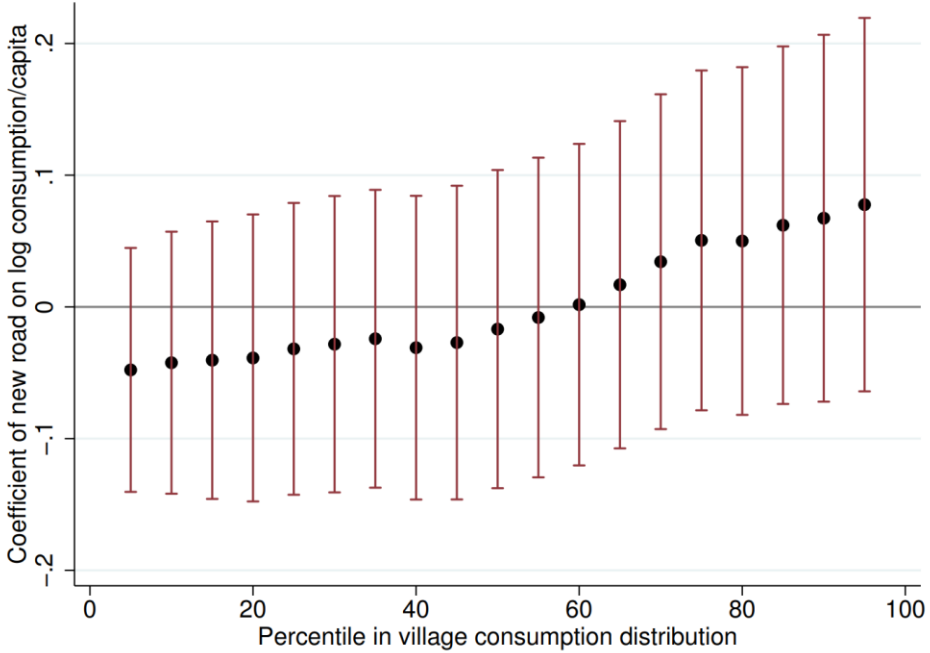
	Consumption per capita (log)	Poverty rate	Night lights (log)	Share of HH earning $\geq$ INR 5k
New road	0.022	-0.010	0.033	-0.001
	(0.038)	(0.042)	(0.165)	(0.032)
Control group mean	9.571	0.282	1.444	0.147
N	11432	11432	11102	11432
R2	0.41	0.30	0.66	0.25

Panel B. Asset ownership

	Asset index	Solid house	Refrigerator	Vehicle	Phone
New road	0.107	0.033	0.005	-0.001	0.033
	(0.132)	(0.029)	(0.013)	(0.023)	(0.041)
Control group mean	-0.015	0.222	0.036	0.140	0.443
N	11432	11432	11432	11432	11432
R2	0.52	0.67	0.26	0.38	0.48

Distributional Effects on Consumption

Figure 6: Distributional impacts of new road on predicted consumption



Summary of Results

What roads affect

- Increase transportation availability
- Allow for exit from agriculture
- Provide access to external labor markets

What roads don't affect

- Agriculture production
- Village non-farm employment
- Consumption
- Major village economy effects
- **In the short run!**

Robustness

05

Robustness Testing

Placebo Testing

Table A9: First stage and reduced form estimates, main and placebo samples

Panel A. Main sample first stage and reduced form effects

	First stage	Reduced form				
	Road by 2012	Transport	Occupation (ag share)	Firms	Ag production	Consumption
Road priority	0.215	0.088	-0.073	0.060	0.018	0.007
	(0.017)	(0.040)	(0.034)	(0.035)	(0.027)	(0.030)
Control group mean	0.25	0.00	0.00	-0.00	-0.00	0.00
N	11432	11432	11432	10678	11432	11432
R2	0.30	0.20	0.30	0.31	0.54	0.50

Panel B. Placebo sample first stage and reduced form effects

	First stage	Reduced form				
	Road by 2012	Transport	Occupation (ag share)	Firms	Ag production	Consumption
Road priority	-0.002	-0.002	-0.016	0.010	-0.047	-0.013
	(0.017)	(0.060)	(0.039)	(0.040)	(0.032)	(0.035)
Control group mean	0.26	0.44	-0.22	0.23	-0.26	0.33
N	9142	9138	9081	8457	9142	9142
R2	0.35	0.29	0.41	0.49	0.51	0.47

Robustness Testing

Placebo Testing

Bandwidth Testing

Table A10: Impact of new road on indices of major outcomes, by kernel and bandwidth

	Triangular			Rectangular		
	60	80	100	60	80	100
Transport	0.404	0.411	0.401	0.419	0.430	0.307
	(0.208)	(0.188)	(0.172)	(0.205)	(0.182)	(0.154)
	[0.05]	[0.03]	[0.02]	[0.04]	[0.02]	[0.05]
Ag occupation	-0.290	-0.337	-0.332	-0.343	-0.362	-0.260
	(0.181)	(0.162)	(0.148)	(0.176)	(0.157)	(0.133)
	[0.11]	[0.04]	[0.02]	[0.05]	[0.02]	[0.05]
Firms	0.394	0.281	0.235	0.275	0.159	0.172
	(0.177)	(0.158)	(0.144)	(0.172)	(0.153)	(0.131)
	[0.03]	[0.07]	[0.10]	[0.11]	[0.30]	[0.19]
Ag production	0.145	0.093	0.071	0.102	0.080	0.050
	(0.139)	(0.125)	(0.114)	(0.137)	(0.121)	(0.104)
	[0.30]	[0.46]	[0.54]	[0.46]	[0.51]	[0.63]
Consumption	0.112	0.063	0.035	0.098	0.030	-0.023
	(0.154)	(0.138)	(0.126)	(0.149)	(0.133)	(0.112)
	[0.47]	[0.65]	[0.78]	[0.51]	[0.82]	[0.84]
N	[8339]	[11099]	[13871]	[8339]	[11099]	[13871]

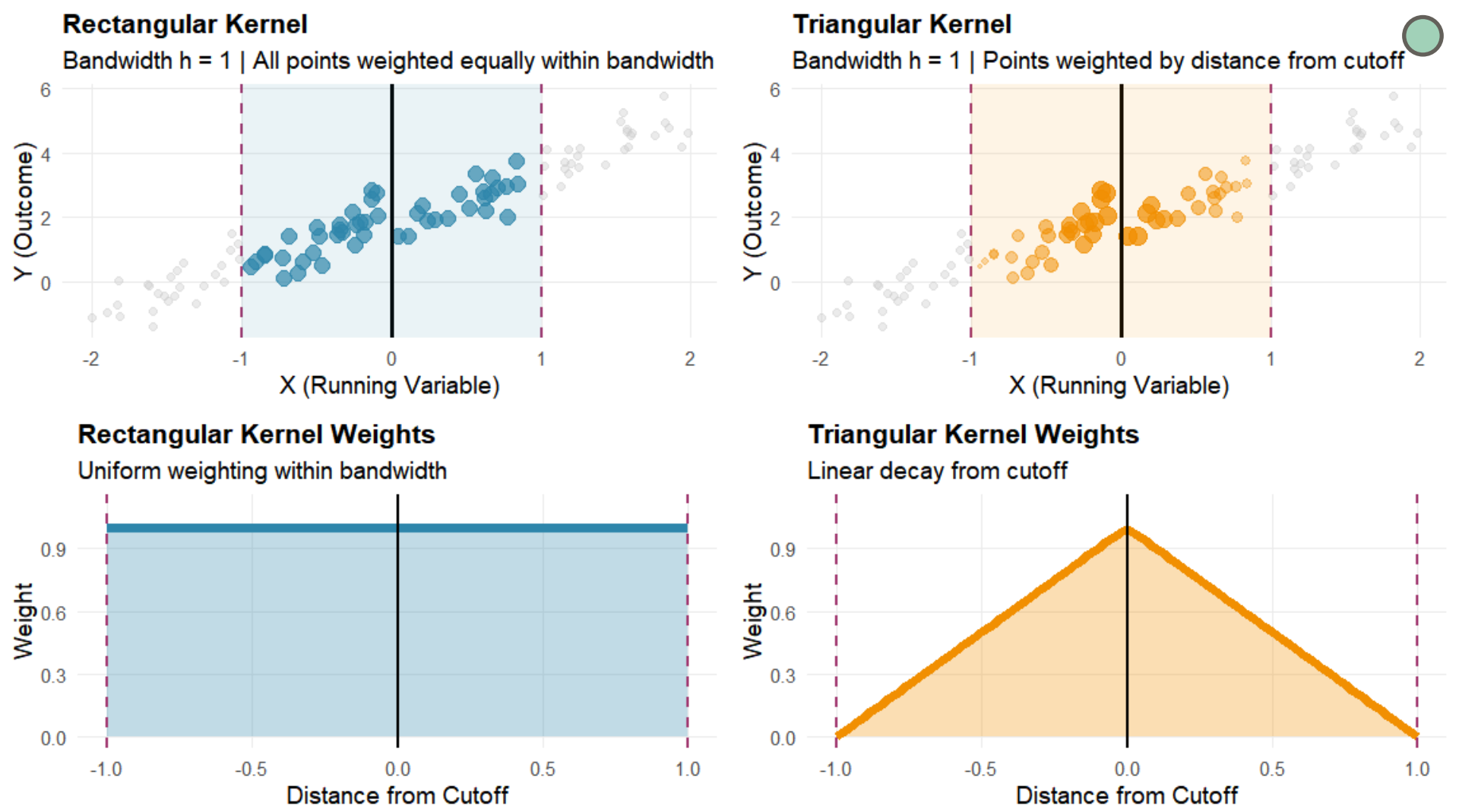

Rectangular Kernel
Bandwidth h = 1 | All points weighted equally within bandwidth
Y (Outcome)
X (Running Variable)
Triangular Kernel
Bandwidth h = 1 | Points weighted by distance from cutoff
Y (Outcome)
X (Running Variable)
Rectangular Kernel Weights
Uniform weighting within bandwidth
Weight
Distance from Cutoff
Triangular Kernel Weights
Linear decay from cutoff
Weight
Distance from Cutoff

Table A10: Impact of new road on indices of major outcomes, by kernel and bandwidth

	Triangular			Rectangular		
	60	80	100	60	80	100
Transport	0.404	0.411	0.401	0.419	0.430	0.307
	(0.208)	(0.188)	(0.172)	(0.205)	(0.182)	(0.154)
	[0.05]	[0.03]	[0.02]	[0.04]	[0.02]	[0.05]
Ag occupation	-0.290	-0.337	-0.332	-0.343	-0.362	-0.260
	(0.181)	(0.162)	(0.148)	(0.176)	(0.157)	(0.133)
	[0.11]	[0.04]	[0.02]	[0.05]	[0.02]	[0.05]
Firms	0.394	0.281	0.235	0.275	0.159	0.172
	(0.177)	(0.158)	(0.144)	(0.172)	(0.153)	(0.131)
	[0.03]	[0.07]	[0.10]	[0.11]	[0.30]	[0.19]
Ag production	0.145	0.093	0.071	0.102	0.080	0.050
	(0.139)	(0.125)	(0.114)	(0.137)	(0.121)	(0.104)
	[0.30]	[0.46]	[0.54]	[0.46]	[0.51]	[0.63]
Consumption	0.112	0.063	0.035	0.098	0.030	-0.023
	(0.154)	(0.138)	(0.126)	(0.149)	(0.133)	(0.112)
	[0.47]	[0.65]	[0.78]	[0.51]	[0.82]	[0.84]
N	[8339]	[11099]	[13871]	[8339]	[11099]	[13871]

Robustness Testing

Migration

Migration Testing

- Check Population Changes
- Check Demographic Changes

Table A11: Impact of new road on population growth, age distribution and gender ratios

Panel A. Population growth (2001-2011)

	Log	Level
New road	-0.024	-9.662
	(0.029)	(20.275)
Control group mean	6.43	653.06
N	11432	11432
R2	0.79	0.83

Panel B. Age group share

	11-20	21-30	31-40	41-50	51-60
New road	-0.004	-0.003	0.002	-0.002	0.002
	(0.005)	(0.004)	(0.004)	(0.004)	(0.003)
Control group mean	0.24	0.19	0.15	0.11	0.07
N	11432	11432	11432	11432	11432
R2	0.22	0.19	0.26	0.38	0.40

Panel C. Male share by age group

	11-20	21-30	31-40	41-50	51-60
New road	-0.010	0.003	0.004	-0.006	0.017
	(0.009)	(0.008)	(0.008)	(0.010)	(0.013)
Control group mean	0.52	0.52	0.51	0.52	0.51
N	11432	11432	11432	11432	11432
R2	0.13	0.19	0.10	0.07	0.05

Robustness Testing

Migration

Spillover

Spillover to Local Villages

Table A14: Spillovers: impact of new road on nearby villages

	Transportation	Ag occupation	Firms	Ag production	Consumption	Unemployment rate
New road	-0.049	-0.001	-0.165	0.036	0.060	-0.007
	(0.135)	(0.132)	(0.141)	(0.100)	(0.114)	(0.009)
p-value	0.72	1.00	0.24	0.72	0.60	0.45
N	11403	11403	11403	11403	11403	11403
R2	0.51	0.52	0.46	0.71	0.65	0.70

So Why Such Small Effects?

Ambiguous Effects

Price effects
vs.
Wage effects

Results Suggest Wage Effects

Workers leave
Only marginal gains

Lack of Complementary Factors

Effective transport
Human capital
Additional infrastructure

Comparison to Prior Literature

Larger Effects Found In

Donaldson 2018, Faber 2014

- Major Highways
- Corridor Infrastructure

Aggarwal 2018

- Same PMGSY program
- District-level observations

Why The Difference?

Selection Bias

Districts building more roads are growing anyway

Villages vs. Districts

Spillovers matter at aggregate level

Feeder Roads vs Highways

Different scale of impact

06

Conclusions

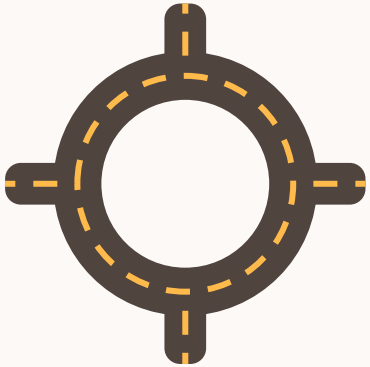
Primary Findings

Bottom Line

Rural roads facilitate labor reallocation,
NOT economic transformation

Insignificant Changes:

Agriculture production,
village firms, or
consumption



Transportation

Increased transportation
access, specifically
government-subsidized

Labor Markets

Increased access to
external labor markets

Contributions

Theoretical

- Even with better market access, remote areas lack complementary factors
- Transport costs necessary but not sufficient for development

Methodological

- First large-scale RD study of rural roads
- Demonstrates power of administrative microdata

Policy Implications

Cost-Benefit

- Cost: $150,000 per village
- Benefit: $4,000/year per village

Roads Are Not Enough

- Need complementary investments
- Education/skills, access to finance, local public goods, clustering of firms, etc.

Policy Implications

More Nuanced View

- Primary benefit: Assume labor market access
- Don’t expect local economic transformation
- May require regional development strategy

Caveats

Limitations

- Short-run effects (4 years average)
- Can't measure prices directly
- Sample limited to certain states

Not Captured Effects

- Long-run growth effects & migration effects
- Health/education access
- Regional equilibrium effects

External Validity

Are Results Generalizable?

- Data is limited to only central and northern India
- Do results generalize to the rest of India?
- What about other developing international regions?

Future Research

Long-run follow-up

Where do workers go?

Complementary interventions

Final Takeaway

Lack of Road Access

Nearly 1 billion people lack road access

Massive Investment

Large amount of investment towards road development, worldwide

Rigorous Evidence

This paper is the first rigorous evidence on village-level impacts

“Even with better market connections, remote areas may continue to lack economic opportunities.”

— Asher and Novosad, 2019

Thanks!